Preference-CFR: Beyond Nash Equilibrium for Better Game Strategies

An Empirical Study of Equilibrium Selection
and Controlled Deviation in Regret Minimization

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The Problem: CFR Picks One Equilibrium

- CFR is the algorithm behind superhuman poker AIs (Pluribus, Libratus).
- Many games have multiple Nash equilibria — all equally "correct".
- Standard CFR picks one on its own. The user has no control.
- Different equilibria = different playing styles (aggressive vs conservative).
- Mention Lee sedol incident. (Wakabayashi & Young, 2024)
- Why play boring equilibrium reaching strategies when with friends?
- **Pref-CFR [Ju et al., ICML 2025] fixes this with two parameters:**

Preference degree δ

Biases toward preferred actions.

With β=0, still converges to exact NE.

$$\sigma^i_{T+1}(I,a) = \begin{cases} \frac{\delta(I,a)\bar{R}^{i,+}_T(I,a)}{\sum_{a\in\mathcal{A}(I)}\delta(I,a)\bar{R}^{i,+}_T(I,a)} & \text{if } \bar{R}^{i,+}_T(I) \neq \mathbf{0} \\ \frac{\delta(I,a)-1}{\sum_{a\in\mathcal{A}^i(I)}\delta(I,a)-1} & \text{otherwise.} \end{cases} \tag{13}$$

Vulnerability degree β

Accepts ε-NE with ε ≤ β.

More freedom for stylistic expression.

$$\bar{B}^i_T(I,a) = \bar{R}^i_T(I,a) - \beta(I), \tag{15}$$

the strategy for time $T+1$ is determined as follows:

$$\sigma^i_{T+1}(I,a) = \begin{cases} \arg\max_{a\in\mathcal{A}^i} \bar{B}^i_T(I,a) & \text{if } \bar{B}^{i,+}_T(I,a) \neq 0 \\ \frac{\delta(I,a)-1}{\sum_{a\in\mathcal{A}^i(I)}\delta(I,a)-1} & \text{otherwise,} \end{cases} \tag{16}$$

Ju et al., Preference-CFR: Beyond Nash Equilibrium for Better Game Strategies, ICML 2025

Our Three Research Questions

RQ1: Implicit Selection

Does standard CFR always converge to the same equilibrium, regardless of initialization?

Game: Kuhn poker
Trials: 30 runs

RQ2: Effect of δ

How strongly does δ shift which equilibrium is selected? Can it cover the full range?

Game: Kuhn poker
Trials: 5 settings × 10

RQ3: Effect of β

How does increasing β affect strategy quality and playing style?

Game: Leduc poker
Trials: 5 β values × 5

It’s good to know how to play these games !!

Kuhn Poker

Rules:

- **Deck:** Only 3 cards are used (Jack, Queen, King).
- **Setup:** 2 players post a blind bet and are each dealt exactly 1 private card.
- **Actions:** Players take turns choosing to either Bet or Pass (check).
- **Response:** If a player bets, the opponent must either Call or Fold.
- **Showdown:** If both players check or a bet is called, hands are revealed. The highest card wins the pot (K > Q > J).

Information Sets:

- **Complexity:** Extremely small game tree.
- **Total Count:** There are only 12 total information sets in the entire game

Leduc Poker

Rules:

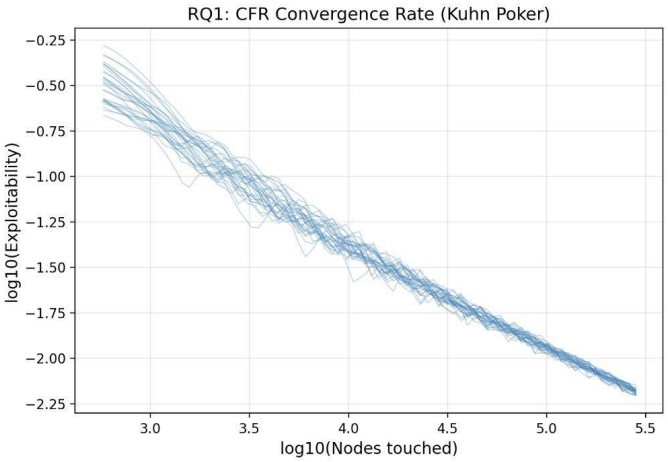
- **Deck:** 6 cards total, comprising the J, Q, and K across two suits (e.g., Spades and Hearts).
- **Setup:** 2 players post blinds and are each dealt 1 private hole card.
- **Round 1:** Players act with options to Bet, Check, Call, or Fold.
- **Round 2:** A single community card is dealt face up, followed by a second round of betting with the same action options (usually with fixed limits).
- **Showdown:** A pair (matching your hole card to the community card) is the strongest hand. If no pair exists, the highest card wins, with suit tie-breakers (e.g., Spades beat Hearts) if ranks match.

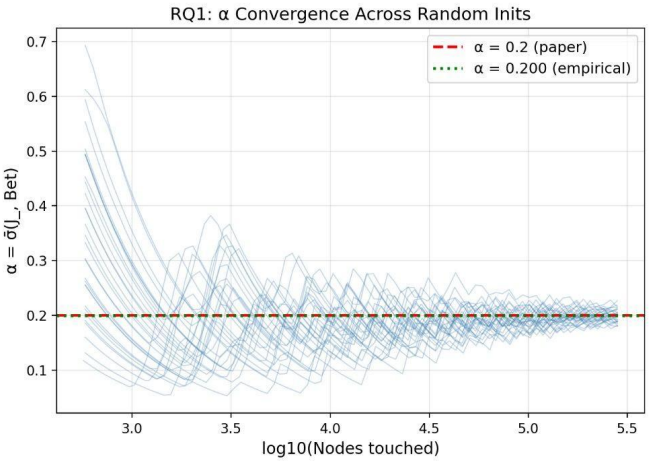
Information Sets:

- **Complexity:** Intermediate complexity bridge between Kuhn and Texas Hold'em.
- **Total Count:** Contains approximately 1,000 information sets.

RQ1: CFR Always Converges to α ≈ 0.200

α = Pr[Bet | Jack] — the bluffing frequency. It identifies which NE we are at.





30 trials

Mean α = 0.200

Std = 0.012

Paper: α = 0.2

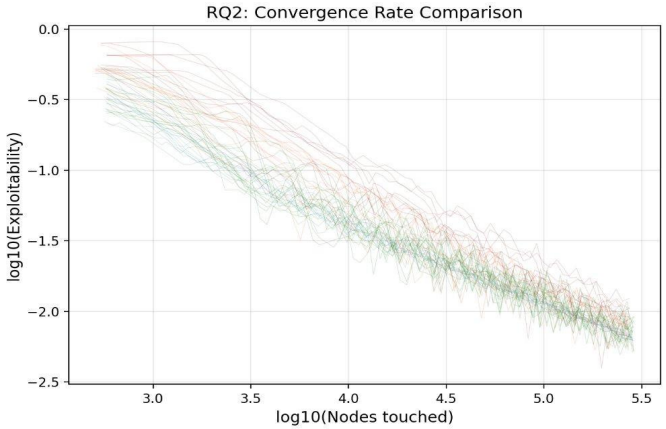
Supports

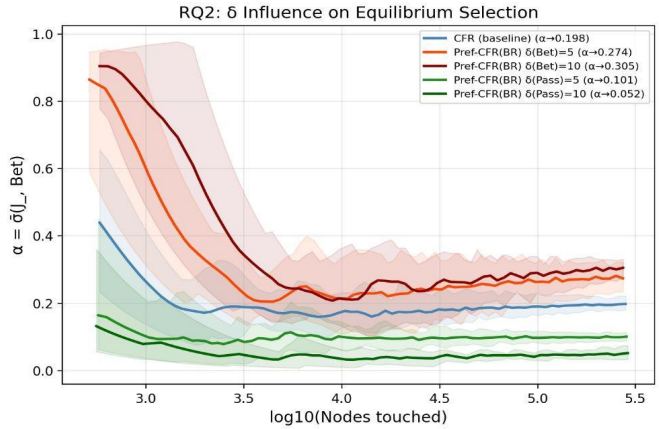
"Conjecture 3.1. In a two-player zero-sum game, if the set of Nash equilibrium ΣNE forms a convex polyhedron, then for any initial strategy σt=0 , the RM iteration converges to a unique fixed point σRM ∈ ΣNE ."

Our replication: 30 independent runs on Kuhn poker, 5000 iterations each. All converge to α ≈ 0.2 regardless of initialization.

Zinkevich et al., Regret minimization in games with incomplete information, NeurIPS 2007

RQ2: δ Gives Monotonic Control Over Equilibrium





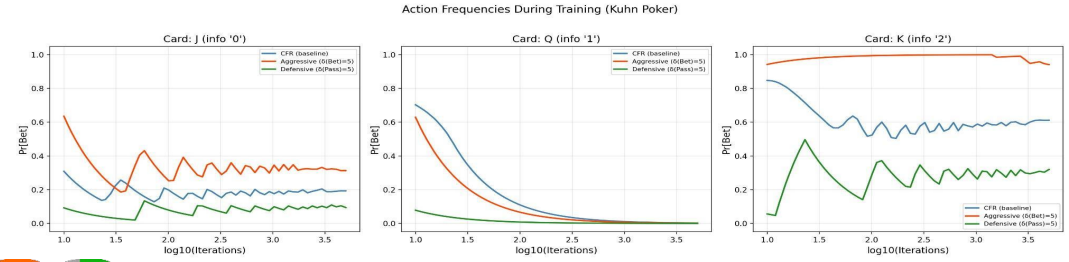
Key findings

δ(Bet)=10: α→0.305
δ(Pass)=10: α→0.052

Spans 76% of feasible range

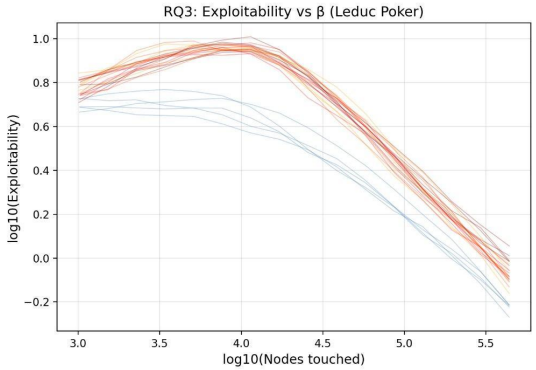
Convergence speed barely affected

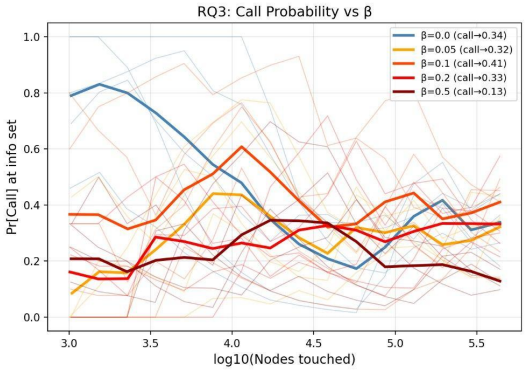
Our replication (not in base paper at this granularity): 10 trials per setting, β=0. Diminishing returns: doubling δ from 5→10 gives only 41% more shift.

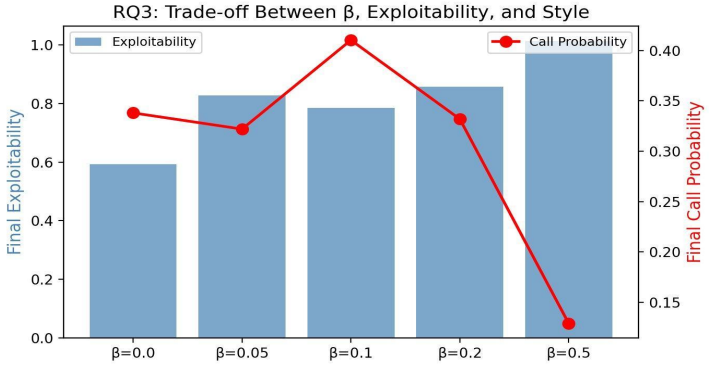


Intersting finding: Queen locked at Pr[Bet]=0 regardless of δ (equilibrium constraint). Only Jack and King respond.

RQ3: Exploitability vs β in Leduc Poker







Important caveat: Baseline (β=0) exploitability is 0.593 — algorithm has NOT converged. Paper used 200× our node budget.

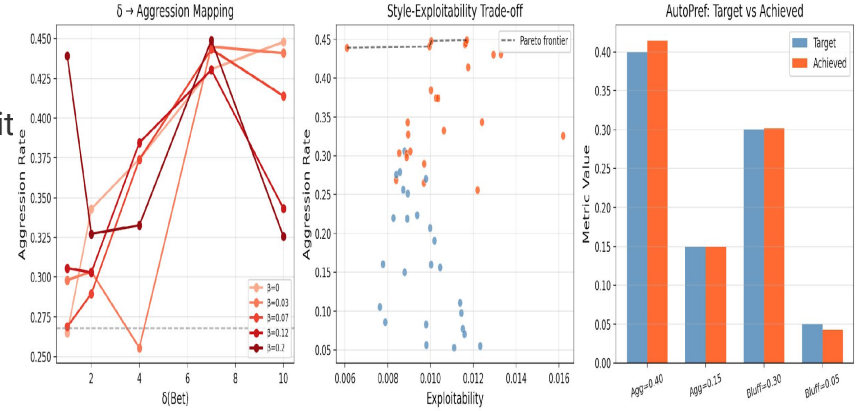
```
ε_observed = ε_train(T) + ε_β
```

Training error dominates → cannot cleanly test ε ≤ β bound at this budget.

Unexpected: Pr[Call] is non-monotonic — peaks at β=0.1, drops at β=0.5. Suggests complex δ×β interaction.

Our Contribution: AutoPref Framework

Open problem from paper (Section 4.2): *"Translating macro-level style characteristics recognized by the public into parameter settings is challenging."*



Our insight (from RQ2):

The δ → style mapping is smooth and monotonic. Pre-compute it once, then look up.

Phase 1 (Calibrate): Run Pref-CFR for a grid of (δ, β) values. Record style metrics + exploitability. (50 configs, 2 min)

Phase 2 (Query): User says "aggression = 40%" → nearest-neighbor lookup → returns δ=10, β=0.07.

AutoPref Results (Kuhn poker, ε_max = 0.02):

"Aggressive" → Agg=0.40 target → achieved 0.414 (expl: 0.012)

"Conservative" → Agg=0.15 target → achieved 0.150 (expl: 0.009)

"Heavy bluffer" → Bluff=0.30 target → achieved 0.302 (expl: 0.013)

"Honest player" → Bluff=0.05 target → achieved 0.043 (expl: 0.011)

Limitations and Future Work

Limitations

- RQ3 under-converged: 500K nodes vs paper's 10^8
- Toy games only (Kuhn, Leduc) — no Texas Hold'em
- Track single parameter (α), not full strategy vector
- PrefCFR(RM) not tested (paper says it is worse)
- No formal statistical significance tests

Future Work

- Scale RQ3 to 10^8 nodes (~1–4 hrs on our hardware)
- Characterize why CFR selects α=0.2 specifically
- Joint (δ, β) grid search on Leduc poker
- Extend AutoPref to larger games via abstraction
- Test with Deep CFR (neural network approx.)

Summary

Base paper contributions (Ju et al., ICML 2025):

Pref-CFR algorithm with δ and β parameters; convergence proofs via Blackwell approachability; Texas Hold'em experiments showing aggressive and loose-passive styles.

Our contributions (replication + new work):

1. Confirmed Conjecture 3.1: $\alpha \to 0.200 \pm 0.012$ (30 trials, Kuhn poker)
2. Showed δ spans 76% of equilibrium range with diminishing returns
3. Identified that $\varepsilon \leq \beta$ bound is loose at finite compute; non-monotonic Pr[Call]
4. Per-card analysis: δ only affects unconstrained information sets (new finding)
5. **AutoPref: automatic parameter tuning from style targets (novel framework)**

Code: github.com/numairsayed/pref-cfr

Thank you!

References.bib

1) *Ju et al., Preference-CFR: Beyond Nash Equilibrium for Better Game Strategies, ICML 2025*
2) *Zinkevich et al., Regret minimization in games with incomplete information, NeurIPS 2007*